

## FARO EVANGELISTA, Chile

WMO: 859300

Lat: 52.387S

Lon: 75.097W

Elev: 52

StdP: 100.70

Time Zone: -3.00 (W03)

Period: 86-09

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 0.9        | 1.7 | -2.5                          | 3.1 | 1.9  | -1.5 | 3.3 | 2.3  | 30.8                  | 7.2  | 26.3 | 7.2  | 10.1                  | 180  | 1.163 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 2.5                    | 13.0            | 11.8 | 12.3 | 11.3 | 11.8 | 10.8 | 12.2                | 12.7 | 11.6 | 12.1 | 11.0 | 11.5 | 11.6                 | 0    |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 11.9                            | 8.7 | 12.4 | 11.2 | 8.4 | 11.8 | 10.7 | 8.0 | 11.3 | 34.9          | 12.7 | 33.5 | 12.1 | 32.1 | 11.6 | 15.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 26.8              | 23.5 | 21.3 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | -0.7                       | 13.1 | 0.9                | 1.2 | -1.4                                               | 13.9 | -1.9       | 14.6 | -2.4       | 15.3 | -3.0       | 16.2 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.3    | 8.9   | 9.6   | 9.1   | 8.2   | 7.2   | 5.8   | 5.1   | 5.3   | 5.6   | 6.5   | 7.5   | 8.5   |
|                                                                        | DBStd       | 2.18   | 1.16  | 1.51  | 1.57  | 1.61  | 1.80  | 1.92  | 1.82  | 1.54  | 1.55  | 1.41  | 1.36  | 1.27  |
|                                                                        | HDD10.0     | 1030   | 36    | 22    | 37    | 57    | 90    | 126   | 151   | 146   | 133   | 107   | 77    | 48    |
|                                                                        | HDD18.3     | 4040   | 292   | 244   | 286   | 304   | 347   | 376   | 409   | 404   | 383   | 366   | 326   | 303   |
|                                                                        | CDD10.0     | 32     | 3     | 12    | 9     | 3     | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 3     |
|                                                                        | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 10.7   | 12.0  | 10.3  | 11.2  | 11.3  | 9.9   | 10.0  | 10.9  | 10.3  | 9.9   | 10.6  | 10.9  | 11.2  |
| (15) Precipitation                                                     | PrecAvg     | 4166   | 385   | 301   | 409   | 329   | 308   | 315   | 304   | 286   | 249   | 306   | 324   | 329   |
|                                                                        | PrecMax     | 5607   | 814   | 479   | 702   | 606   | 637   | 966   | 798   | 684   | 494   | 770   | 694   | 589   |
|                                                                        | PrecMin     | 3146   | 128   | 99    | 186   | 66    | 43    | 56    | 126   | 118   | 36    | 72    | 60    | 84    |
|                                                                        | PrecStd     | 696    | 180   | 105   | 121   | 131   | 166   | 198   | 162   | 132   | 101   | 146   | 153   | 148   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 12.7  | 14.5  | 13.4  | 12.5  | 12.3  | 10.7  | 9.2   | 9.6   | 9.7   | 10.6  | 12.6  | 13.0  |
|                                                                        |             | MCWB   | 11.5  | 13.0  | 12.1  | 11.4  | 10.7  | 9.8   | 8.4   | 8.6   | 8.4   | 8.9   | 11.1  | 11.2  |
|                                                                        | 2%          | DB     | 11.6  | 13.2  | 12.5  | 11.5  | 11.4  | 10.0  | 8.8   | 8.7   | 8.9   | 9.5   | 10.9  | 11.8  |
|                                                                        |             | MCWB   | 10.7  | 12.2  | 11.5  | 10.6  | 10.4  | 9.3   | 8.0   | 8.0   | 7.9   | 8.4   | 9.8   | 10.5  |
|                                                                        | 5%          | DB     | 11.1  | 12.5  | 12.0  | 11.0  | 10.4  | 9.1   | 8.2   | 8.1   | 8.3   | 9.0   | 10.1  | 11.0  |
|                                                                        |             | MCWB   | 10.3  | 11.7  | 11.1  | 10.2  | 9.6   | 8.3   | 7.5   | 7.4   | 7.4   | 8.2   | 9.2   | 9.9   |
|                                                                        | 10%         | DB     | 10.6  | 11.9  | 11.4  | 10.5  | 9.8   | 8.4   | 7.7   | 7.5   | 7.8   | 8.6   | 9.5   | 10.4  |
|                                                                        |             | MCWB   | 9.9   | 11.1  | 10.5  | 9.7   | 9.0   | 7.6   | 6.9   | 6.7   | 7.0   | 7.8   | 8.6   | 9.4   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 12.0  | 13.4  | 12.4  | 11.7  | 11.2  | 10.2  | 8.8   | 9.0   | 9.1   | 9.3   | 11.2  | 11.6  |
|                                                                        |             | MCDB   | 12.4  | 13.9  | 12.9  | 12.3  | 11.7  | 10.4  | 9.1   | 9.4   | 9.4   | 9.9   | 12.1  | 12.6  |
|                                                                        | 2%          | WB     | 11.0  | 12.6  | 11.8  | 10.9  | 10.5  | 9.4   | 8.2   | 8.2   | 8.2   | 8.8   | 10.1  | 10.7  |
|                                                                        |             | MCDB   | 11.2  | 13.1  | 12.3  | 11.3  | 11.1  | 9.8   | 8.6   | 8.7   | 8.6   | 9.2   | 10.6  | 11.5  |
|                                                                        | 5%          | WB     | 10.5  | 12.0  | 11.3  | 10.4  | 9.8   | 8.5   | 7.7   | 7.5   | 7.6   | 8.4   | 9.4   | 10.1  |
|                                                                        |             | MCDB   | 10.8  | 12.4  | 11.8  | 10.8  | 10.3  | 9.0   | 8.2   | 8.0   | 8.1   | 8.9   | 9.9   | 10.7  |
|                                                                        | 10%         | WB     | 10.1  | 11.2  | 10.7  | 9.9   | 9.1   | 7.7   | 7.0   | 6.9   | 7.1   | 8.0   | 8.8   | 9.7   |
|                                                                        |             | MCDB   | 10.5  | 11.7  | 11.3  | 10.4  | 9.6   | 8.3   | 7.6   | 7.3   | 7.7   | 8.5   | 9.3   | 10.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 2.2   | 2.5   | 2.4   | 2.2   | 2.3   | 2.3   | 2.2   | 2.1   | 2.3   | 2.4   | 2.5   | 2.3   |
|                                                                        |             | MCDBR  | 2.7   | 3.7   | 3.0   | 2.7   | 3.2   | 2.8   | 2.9   | 2.5   | 2.7   | 2.9   | 3.4   | 3.2   |
|                                                                        |             | MCWBR  | 2.7   | 3.5   | 2.7   | 2.8   | 2.9   | 2.8   | 3.0   | 2.5   | 2.5   | 2.7   | 3.1   | 2.7   |
|                                                                        | 5% WB       | MCDBR  | 2.4   | 3.5   | 2.8   | 2.6   | 3.1   | 2.7   | 2.9   | 2.3   | 2.5   | 2.7   | 3.2   | 2.8   |
|                                                                        |             | MCWBR  | 2.7   | 3.4   | 2.9   | 2.9   | 3.1   | 2.9   | 3.1   | 2.5   | 2.6   | 2.7   | 3.1   | 2.7   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.361 | 0.344 | 0.335 | 0.345 | 0.324 | 0.319 | 0.327 | 0.341 | 0.336 | 0.343 | 0.342 | 0.353 |
|                                                                        | taud        |        | 2.450 | 2.517 | 2.533 | 2.377 | 2.311 | 2.228 | 2.265 | 2.328 | 2.457 | 2.459 | 2.489 | 2.467 |
|                                                                        | Ebn at Noon |        | 941   | 923   | 870   | 736   | 621   | 540   | 600   | 724   | 850   | 913   | 956   | 959   |
|                                                                        | Edh at Noon |        | 111   | 96    | 83    | 76    | 60    | 54    | 61    | 78    | 88    | 101   | 106   | 111   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 3.85  | 3.31  | 2.22  | 1.28  | 0.74  | 0.48  | 0.60  | 1.17  | 2.14  | 3.20  | 3.87  | 4.23  |
|                                                                        | RadStd      |        | 0.29  | 0.34  | 0.18  | 0.12  | 0.08  | 0.07  | 0.07  | 0.10  | 0.24  | 0.29  | 0.34  | 0.35  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page